1.2.7 Selman Abraham Waksman (1888-1973)

Waksman was from Rutger University, USA (Figure 1.11). His research was largely on soil microorganisms. He showed antimicrobial activity of streptomyces that led to the discovery of Streptomycin and several other antibiotics.

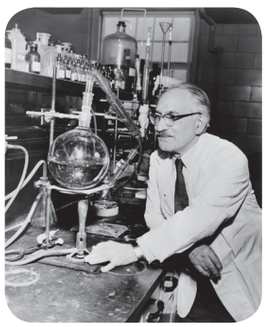


Figure 1.11: Selman Abraham Waksman (1888-1973)

Waksman and his co-workers isolated Actinomycin in 1940, Streptothrecin in 1942, Streptomycin in 1943, and Neomycin in 1949.

Streptomycin is produced by *Streptomyces griseus*. It is a secondary metabolite produced by *Streptomyces griseus* which is not required for its growth but may help it to compete with other bacteria for food and space in the environment. Streptomycin is used in the treatment of tuberculosis. Waksman got Nobel Prize in 1952. for his work on Streptomycin



Antibiotics are usually not effective for sore throats and common colds. They are commonly caused by viruses rather than bacteria. Taking antibiotics for such illnesses is considered more harmful than beneficial.

1.3 Branches of Microbiology

Microbiology can be classified into Pure and Applied Microbiology. Pure Microbiology is classified based on taxonomical and integrative characteristics. Table 1.1 shows various branches of microbiology.

Table 1.1: Branches of Microbiology

Based on Taxonomical characteristics	
Bacteriology	The study of bacteria
Mycology	The study of fungi
Protozoology	The study of protozoa
Based on Taxonomical characteristics	
Phycology (or algology)	The study of algae
Parasitology	The study of parasites
Immunology	The study of the immune system
Virology	The study of viruses
Nematology	The study of the nematodes

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